

MATH 345 Skeleton Notes

Unit 5: Optimization and training

Section 5.2: Directional derivatives and gradient descent

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Directional derivatives

Activity

Find the system of equations satisfied by the critical points of the function

$$f(x, y, z) = x^2 e^{yz} + y \sin(xz) + z \cos(xy).$$

We calculate that

$$f_x(x, y, z) = 2xe^{yz} + yz \cos(xz) - yz \sin(xy),$$

that

$$f_y(x, y, z) = x^2 z e^{yz} + \sin(xz) - xz \sin(xy)$$

and that

$$f_z(x, y, z) = x^2 y e^{yz} + xy \cos(xz) + \cos(xy).$$

So the critical points of f are the solutions to the system of equations

$$\begin{cases} 2xe^{yz} + yz \cos(xz) - yz \sin(xy) = 0, \\ x^2 z e^{yz} + \sin(xz) - xz \sin(xy) = 0, \\ x^2 y e^{yz} + xy \cos(xz) + \cos(xy) = 0. \end{cases}$$

Using the calculus we have developed, it is usually straightforward to write the equations that critical points must satisfy. The activity also shows the problem: those equations may be nonlinear and difficult or impossible to solve exactly. This is different from the linear systems in Unit 2, where Gaussian elimination gave a systematic exact method.

For such problems, one often uses numerical methods. Gradient descent starts with an initial guess $\mathbf{x}_0$ and produces iterates

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

The positive scalar α is the step size. The minus sign moves in a direction where f initially decreases. The method does not automatically converge; the function, starting point, and step size matter.

To discuss this further, we must introduce more general kinds of derivatives. Recall that the partial derivatives of a function f are defined as

$$\frac{\partial f}{\partial x_i}(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{e}_i) - f(\mathbf{a})}{h}.$$

By replacing $\mathbf{e}_i$ by a more general unit vector, we obtain the concept of a *directional derivative*.

Definition Let f be a scalar-valued function. Then if $\mathbf{v}$ is a unit vector, the

directional derivative of f in the direction of the vector $\mathbf{v}$, denoted $D_{\mathbf{v}}f$, is defined as

$$D_{\mathbf{v}}f(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{v}) - f(\mathbf{a})}{h},$$

provided the limit exists.

The directional derivative represents the rate of change of the function f at the point $\mathbf{a}$ in the direction of the unit vector $\mathbf{v}$. For a function $f(x, y)$ of two variables, the directional derivative $D_{\mathbf{v}}f(\mathbf{a})$ is the slope of the tangent line to the curve obtained by intersecting the surface $z = f(x, y)$ with the plane containing both the line passing through $\mathbf{a}$ with direction vector $\mathbf{v}$, and the z axis.

Remark Partial derivatives are a special case of directional derivatives, i.e. we have

$$\frac{\partial f}{\partial x_i} = D_{\mathbf{e}_i}f.$$

Despite partial derivatives being a special case of directional derivatives, in most circumstances we can compute directional derivatives using partial derivatives, or equivalently, the gradient.

Theorem Suppose f is a function differentiable at a point $\mathbf{a}$. Then for each unit vector $\mathbf{v}$, $D_{\mathbf{v}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}$.

Proof We apply the chain rule. We note that $D_{\mathbf{v}}f(\mathbf{a})$ is equal to the derivative of the function $g(t) = f(\mathbf{a} + t\mathbf{v})$ at $t = 0$ (both derivatives are defined as the same limits). If we write $\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}$, then $g = f \circ \mathbf{c}$. But the chain rule (Chain Rule for One Independent Variable) implies that

$$g'(0) = (\nabla f)(\mathbf{c}(0)) \cdot \mathbf{c}'(0) = (\nabla f)(\mathbf{a}) \cdot \mathbf{v}.$$

The theorem greatly simplifies the computations of directional derivatives.

Activity: Dot product controls direction

Suppose

$$\nabla f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Compute $D_{\mathbf{v}}f(\mathbf{a})$ for each unit vector below:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{v}_4 = -\frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

1. Which listed direction gives the largest directional derivative?
2. Which listed direction gives the smallest directional derivative?
3. How is this the Unit 1 dot product appearing again?

Tags. [U5-L02, U1-L01 | C+R | Core]

Since

$$D_{\mathbf{v}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v},$$

we compute

$$D_{\mathbf{v}_1}f(\mathbf{a}) = 3, \quad D_{\mathbf{v}_2}f(\mathbf{a}) = 4.$$

Also,

$$D_{\mathbf{v}_3}f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \frac{9 + 16}{5} = 5,$$

and

$$D_{\mathbf{v}_4}f(\mathbf{a}) = -5.$$

The largest listed directional derivative occurs in direction $\mathbf{v}_3$, the direction of the gradient. The smallest occurs in direction $\mathbf{v}_4$, the direction opposite the gradient. This is the Unit 1 dot product measuring alignment: larger positive alignment gives larger directional derivative.

Activity

Let $f(x, y) = x^2y + y^2$.

Compute the directional derivative $D_{\mathbf{v}}f(1, 2)$, where $\mathbf{v} = (1/\sqrt{2}, 1/\sqrt{2})$, and $f(x, y) = x^2y + y^2$.

We begin by computing the gradient of f at $(1, 2)$. We have

$$f_x(x, y) = 2xy \quad \text{and} \quad f_y(x, y) = x^2 + 2y.$$

So $\nabla f(x, y) = (2xy, x^2 + 2y)$, and in particular

$$\nabla f(1, 2) = \begin{bmatrix} 2(1)(2) \\ (1)^2 + 2(2) \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

So

$$D_{\mathbf{v}}f(1, 2) = \nabla f(1, 2) \cdot \mathbf{v} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = 9/\sqrt{2}.$$

Activity

Let $f(x, y) = x^2y + y^2$.

Find the directional derivative of f at the point $(-1, 3)$ in the direction of the vector $(-1, \sqrt{3})$. That is, compute $D_{\mathbf{v}}f(-1, 3)$, where $\mathbf{v}$ is the unit vector in the direction of $(-1, \sqrt{3})$.

The vector

$$\begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}$$

has norm

$$\sqrt{(-1)^2 + (\sqrt{3})^2} = 2.$$

So the unit vector in this direction is

$$\mathbf{v} = \frac{1}{2} \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix} = \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix}.$$

For

$$f(x, y) = x^2y + y^2,$$

we have

$$\nabla f(x, y) = \begin{bmatrix} 2xy \\ x^2 + 2y \end{bmatrix}.$$

Thus

$$\nabla f(-1, 3) = \begin{bmatrix} -6 \\ 7 \end{bmatrix}.$$

Therefore

$$D_{\mathbf{v}}f(-1, 3) = \nabla f(-1, 3) \cdot \mathbf{v} = \begin{bmatrix} -6 \\ 7 \end{bmatrix} \cdot \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix} = 3 + \frac{7\sqrt{3}}{2}$$

Activity

Use the gradient of the function $f(x, y) = x^3y$ to find the directional derivative of f at $(2, 1)$ in the direction of the unit vector which we placed at $(2, 1)$ points in the direction of the point $(3, 5)$.

The unit vector pointing in the direction from $(2, 1)$ to $(3, 5)$ is the unit vector pointing in the same direction as the vector $(3 - 2, 5 - 1) = (1, 4)$. This vector has magnitude $\sqrt{1^2 + 4^2} = \sqrt{17}$, so the unit vector pointing in this direction is equal to

$$\mathbf{v} = \begin{bmatrix} 1/\sqrt{17} \\ 4/\sqrt{17} \end{bmatrix}.$$

Now

$$\nabla f(x, y) = \begin{bmatrix} 3x^2y \\ x^3 \end{bmatrix},$$

and in particular,

$$\nabla f(2, 1) = \begin{bmatrix} 3(2)^2(1) \\ (2)^3 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix}.$$

So

$$D_{\mathbf{v}}f(2, 1) = \nabla f(2, 1) \cdot \mathbf{v} = \begin{bmatrix} 12 \\ 8 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{17} \\ 4/\sqrt{17} \end{bmatrix} = \frac{12 + 32}{\sqrt{17}} = \frac{44}{\sqrt{17}}.$$

Activity

Let $f(x, y) = x^2 e^y$. Find the directional derivative of f at $(2, 0)$ in the direction pointing from $(2, 0)$ to $(3, 0)$.

The unit vector is $(1, 0) = \mathbf{e}_1$, and

$$D_{\mathbf{e}_1} f(x, y) = f_x(x, y) = 2xe^y.$$

So in particular

$$D_{\mathbf{e}_1} f(2, 0) = 2(2)e^{(0)} = 4.$$

Theorem Suppose that a function f is differentiable at a point $\mathbf{a}$.

- If $\nabla f(\mathbf{a}) = 0$, then $D_{\mathbf{v}} f(\mathbf{a}) = 0$ for any unit vector $\mathbf{v}$.
- If $\nabla f(\mathbf{a}) \neq 0$, then $D_{\mathbf{v}} f(\mathbf{a})$ is maximized, among unit vectors $\mathbf{v}$, when

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

It is minimized when

$$\mathbf{v} = -\frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

The maximum value is $\|\nabla f(\mathbf{a})\|$, and the minimum value is $-\|\nabla f(\mathbf{a})\|$.

Proof We have already verified the theorem when $\nabla f(\mathbf{a}) = \mathbf{0}$ in the theorem. When $\nabla f(\mathbf{a}) \neq \mathbf{0}$, the theorem and the Cauchy-Schwarz Inequality The Cauchy-Schwarz inequality says that

$$|D_{\mathbf{u}}f(\mathbf{a})| = |\nabla f(\mathbf{a}) \cdot \mathbf{u}| \leq \|\nabla f(\mathbf{a})\| \|\mathbf{u}\| = \|\nabla f(\mathbf{a})\|,$$

where the inequality is an *equality* precisely when $\mathbf{u}$ either points in the direction of $\nabla f(\mathbf{a})$, or points in the opposite of this direction. But when $\mathbf{u}$ points in the direction of $\nabla f(\mathbf{a})$, i.e. when $\mathbf{u} = \nabla f(\mathbf{a})/\|\nabla f(\mathbf{a})\|$, we have

$$\begin{aligned} D_{\mathbf{u}}f(\mathbf{a}) &= \nabla f(\mathbf{a}) \cdot \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|} \\ &= \frac{\nabla f(\mathbf{a}) \cdot \nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|} = \frac{\|\nabla f(\mathbf{a})\|^2}{\|\nabla f(\mathbf{a})\|} = \|\nabla f(\mathbf{a})\|. \end{aligned}$$

Conversely, one can check that when $\mathbf{u} = -\nabla f(\mathbf{a})/\|\nabla f(\mathbf{a})\|$, then

$$D_{\mathbf{u}}f(\mathbf{a}) = -\|\nabla f(\mathbf{a})\|.$$

So we see that these two vectors are precisely the maximizer and minimizer of the directional derivatives.

Activity

Find the direction for which the directional derivative of $f(x, y) = 3x^2 - 4xy + 2y^2$ at $(-2, 3)$ is a maximum. What is the maximum value of the directional derivative?

First, we calculate that

$$\nabla f(x, y) = \begin{bmatrix} 6x - 4y \\ -4x + 4y \end{bmatrix}.$$

So

$$\nabla f(-2, 3) = \begin{bmatrix} 6(-2) - 4(3) \\ -4(-2) + 4(3) \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix}.$$

the theorem tells us that the directional derivative $D_{\mathbf{v}}f$ is maximized when $\mathbf{v}$ is the unit vector pointing in the direction of $(-24, 20)$. Since $(-24, 20)$ has magnitude $\sqrt{(-24)^2 + (20)^2} = \sqrt{976}$, we see the direction which maximizes the directional derivative is

$$\mathbf{v} = \begin{bmatrix} -24/\sqrt{976} \\ 20/\sqrt{976} \end{bmatrix},$$

and the value of the directional derivative in this direction is equal to the magnitude of the gradient, which is

$$\sqrt{976}.$$

Gradient descent

Remark We now return to the gradient descent algorithm, a method for trying to approximate the local minima of a function f . The algorithm works as follows:

1. Start at an initial point $\mathbf{x}_0$.
2. Compute the gradient $\nabla f(\mathbf{x}_0)$.
3. Move in the direction of the negative gradient: set $\mathbf{x}_1 = \mathbf{x}_0 - \alpha_0 \nabla f(\mathbf{x}_0)$, where $\alpha_0 > 0$ is a step size (or learning rate) parameter.
4. Repeat steps 2-3 with step sizes $\alpha_1, \alpha_2, \dots$ to successively define points $\mathbf{x}_2 = \mathbf{x}_1 - \alpha_1 \nabla f(\mathbf{x}_1)$, $\mathbf{x}_3 = \mathbf{x}_2 - \alpha_2 \nabla f(\mathbf{x}_2)$, and so on, with the hope that these points converge to a local minimum.

Intuitively, gradient descent works because each successive value $\mathbf{x}_k$ is obtained by shifting the previous value $\mathbf{x}_{k-1}$ in the direction in which the function f decreases the fastest. The algorithm is likely to find local minima of a function f , although a rigorous analysis of this algorithm is quite involved.

Activity: Reading one gradient descent update

Assume `grad_f(x)` computes $\nabla f(\mathbf{x})$. Consider the line of code

```
x = x - alpha * grad_f(x)
```

1. What mathematical update rule does this line represent?
2. Which quantity is the learning rate?
3. Why must `grad_f(x)` have the same shape as `x`?
4. What would change if the minus sign were a plus sign?

Tags. [U5-L03 | C+T | Core]

The line represents the update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

The learning rate is `alpha`. The vector `grad_f(x)` must have the same shape as `x`, because the update subtracts one vector from another. If the minus sign were a plus sign, the update would move in the direction of steepest increase rather than steepest decrease.

Activity

Find an approximate minimum of the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x^2 - 3xy + 3y^2 + 5y + 2x$$

using gradient descent. Use the starting point $\mathbf{x}_0 = (0, 0)$, a constant step size of 0.1, and stop after computing $\mathbf{x}_3$. You may use a calculator to perform these computations.

First, we calculate the gradient:

$$\nabla f(x, y) = \begin{bmatrix} 2x - 3y + 2 \\ -3x + 6y + 5 \end{bmatrix}$$

Starting at $\mathbf{x}_0 = (0, 0)$, we successively compute that

$$\nabla f(0, 0) = \begin{bmatrix} 2 \\ 5 \end{bmatrix},$$

so

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} -0.2 \\ -0.5 \end{bmatrix}.$$

We then compute that

$$\nabla f(-0.2, -0.5) = \begin{bmatrix} -0.4 - (-1.5) + 2 \\ -3(-0.2) + 6(-0.5) + 5 \end{bmatrix} = \begin{bmatrix} 3.1 \\ 2.6 \end{bmatrix},$$

so

$$\mathbf{x}_2 = \begin{bmatrix} -0.2 \\ -0.5 \end{bmatrix} - 0.1 \begin{bmatrix} 3.1 \\ 2.6 \end{bmatrix} = \begin{bmatrix} -0.51 \\ -0.76 \end{bmatrix}.$$

Finally, we compute that

$$\nabla f(-0.51, -0.76) = \begin{bmatrix} 2(-0.51) - 3(-0.76) + 2 \\ -3(-0.51) + 6(-0.76) + 5 \end{bmatrix} = \begin{bmatrix} 3.26 \\ 1.97 \end{bmatrix}$$

so

$$\mathbf{x}_3 = \begin{bmatrix} -0.51 \\ -0.76 \end{bmatrix} - 0.1 \begin{bmatrix} 3.26 \\ 1.97 \end{bmatrix} = \begin{bmatrix} -0.836 \\ -0.957 \end{bmatrix}.$$

We note that for this function, we can precisely compute that f has a critical point at $(-9, -16/3)$, which by the second derivative test is a local minimum. So $\mathbf{x}_3$ is *not* a good approximation to the local minimum. But if we continued to iterate this algorithm, we would

find that the points $\mathbf{x}_k$ for, say, $k \geq 1000$, are very close to the true minimum of the function.

Remark This algorithm can be rather cumbersome to compute by hand, but if the gradient of a function is easily determined, can be easily done to a large number of iterations on a computer.

Activity

Calculate $D_{\mathbf{u}}f(1, -2, 3)$ in the direction of $\mathbf{v} = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ for the function

$$f(x, y, z) = 5x^2 - 2xy + y^2 - 4yz + z^2 + 3xz.$$

We compute that

$$\|\mathbf{v}\| = \sqrt{(-1)^2 + (2)^2 + (2)^2} = \sqrt{9} = 3$$

So the unit vector in the direction of $\mathbf{v}$ is the vector

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = -\frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}$$

Using the formula for the directional derivative,

we find that

$$\begin{aligned} D_{\mathbf{u}}f(x, y, z) &= \nabla f(x, y, z) \cdot \mathbf{u} \\ &= \begin{bmatrix} 10x - 2y \\ -2x + 2y - 4z \\ -4y + 2z + 3x \end{bmatrix} \cdot \mathbf{u} \\ &= -\frac{10x - 2y}{3} + 2\frac{-2x + 2y - 4z}{3} + 2\frac{-4y + 2z + 3x}{3} \end{aligned}$$

At the point $(1, -2, 3)$, we evaluate that

$$D_{\mathbf{u}}f(1, -2, 3) = (23) \left(-\frac{1}{3}\right) + (-18) \left(\frac{2}{3}\right) + (17) \left(\frac{2}{3}\right) = \frac{-2}{3}$$